Given: Due:	Further Maths (Decision) HW 16	Name:

Q1.

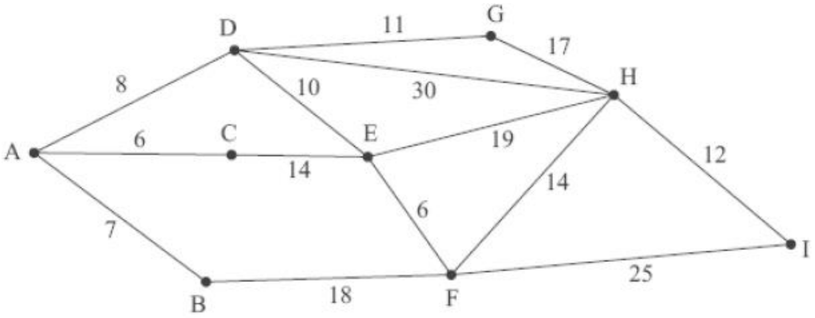


Figure 3

Figure 3 shows a network of roads. The number on each arc represents the length, in km, of that road.

(a) Use Dijkstra's algorithm to find the shortest route from A to I. State your shortest route and its length.

(5)

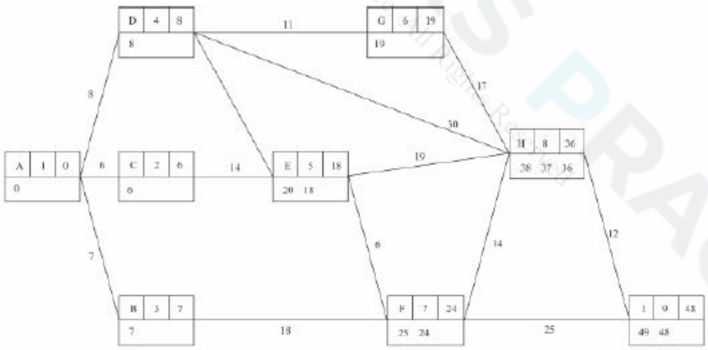
Sam has been asked to inspect the network and assess the condition of the roads. He must travel along each road at least once, starting and finishing at A.

(b) Use an appropriate algorithm to determine the length of the shortest route Sam can travel. State a shortest route.

(4)

(The total weight of the network is 197 km.)

Most candidates found part (a) straightforward and gained full marks, but candidates are reminded that it is the order of the working values that is key for examiners to determine if the algorithm has been applied correctly. Many gained full, or nearly full marks in (b) with errors mostly due to stating an incorrect inspection route (C was frequently omitted). Some wasted time finding the shortest route from A to H from scratch, rather than using their working for part (a), Dijkstra's algorithm finds the length of the shortest route from the start to all intermediate points. Others wasted time finding the weight of the network even though it was given to them.

Question Number	Scheme	Marks
(a)	 Route: ADGHI Length: 48 (km)	M1 A1 A1ft A1 A1ft (5)

(b)	Odd vertices are A and H Attempt to find shortest route from A to H = ADGH New length: 197 + 36 = 233 Route: e.g. ADGHGDACEDHIFHEFBA (18) **Notes:** (a) 1M1: Smaller number replacing larger number in the working values at E or F or H or I. (generous – give bod) 1A1: All values in boxes A to E and G correct 2A1ft: All values in boxes F, H and I correct (ft). Penalise order of labelling just once. 3A1: CAO (not ft) 4A1ft: Follow through from their I value, condone lack of units here. (b) 1B1: A and H identified in some way – allow recovery from M mark. 1M1: Accept, if correct, path, or its length. Accept attempt if finding shortest. 1A1ft: 197 + their shortest A to H (36) 2A1: A correct route.	B1 M1 A1ft A1 (4) **Total 9**

Q2.

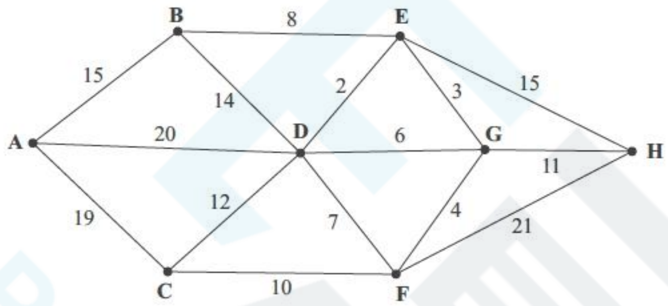


Figure 3

[*The total weight of the network is 167*]

Figure 3 represents a network of paths. The number on each arc gives the time, in minutes, to travel along that path.

(a) Use Dijkstra's algorithm to find the quickest route from A to H. State your quickest route and the time taken.

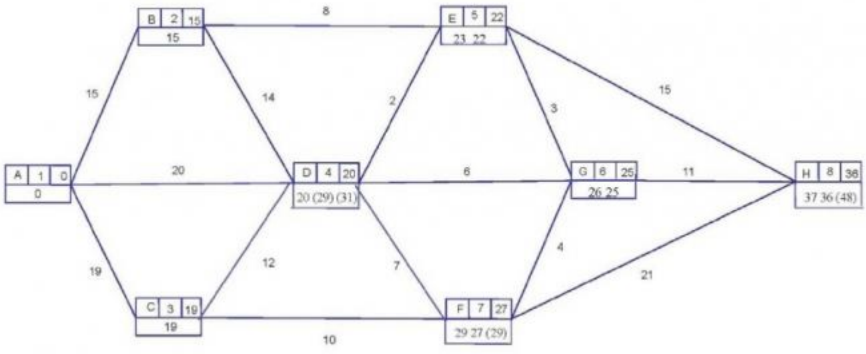
(5)

Kevin must walk along each path at least once and return to his starting point.

(b) Use an appropriate algorithm to find the time of Kevin's quickest possible route, starting and finishing at A. You should make your method and working clear.

(5)

This was generally well done, with most candidates using the boxes sensibly to make their working clear, however some candidates listed their working values in the wrong order and others did not state the correct order of labeling, with F and G often incorrectly ordered. The correct shortest path and its length were frequently seen. The method was usually correct, in part (b), with three pairs being found. However, only the better candidates got all three values correct. Few seemed to make use of the result found in part (a) with 37 frequently stated as the value of AH. This part also saw a proliferation of basic arithmetical errors. Values seen were AB+CH = 43, 44, 45, 46, 59; AC+BH = 42; AH+BC = 55, 59, 61, 62, 63, 64. Some went to the trouble of listing a possible route but then sometimes did not state its length.



Clear method to include at least 1 update (look at E, F, G or H)	M1
BCDE correct	A1
FGH correct	A1ft
Route ADEGH	A1
Total time 36 Minutes	A1ft (5)

Question Number	Scheme	Marks
(b)	Odd nodes are A, B, C, H	M1
	AB + CH = 15 + 25 = 40	A1
	AC + BH = 19 + 22 = 41	A1
	AH + BC = 36 + 22 = 58	A1
	(40 is the shortest, repeating AB and CF + FG + GH)	
	Must be choosing from at least two pairings for this last mark Shortest time = 167 + 40 = 207 minutes. 167 + their shortest	A1ft (5)
		[10]

Q3.

[The total weight of the network is 451]

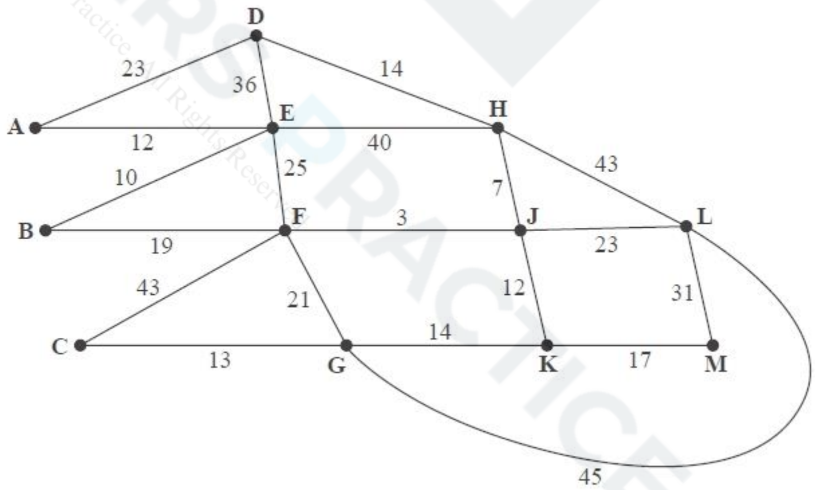


Figure 1 models a network of tracks in a forest that need to be inspected by a park ranger. The number on each arc is the length, in km, of that section of the forest track.

Each track must be traversed at least once and the length of the inspection route must be minimised. The inspection route taken by the ranger must start and end at vertex A.

(a) Use the route inspection algorithm to find the length of a shortest inspection route. State the arcs that should be repeated. You should make your method and working clear.

(5)

(b) State the number of times that vertex J would appear in the inspection route.

(1)

The landowner decides to build two huts, one hut at vertex K and the other hut at a different vertex. In future, the ranger will be able to start his inspection route at one hut and finish at the other. The inspection route must still traverse each track at least once.

(c) Determine where the other hut should be built so that the length of the route is minimised. You must give reasons for your answer and state a possible route and its length.

(4)

Question (a) was generally answered well by most students with the vast majority stating the correct three distinct pairings of the correct four odd nodes. There were a few students who only gave two pairings of the four odd nodes or who gave several pairings but not three distinct pairings. There were however many instances where the totals were incorrect. The majority of such mistakes occurred for the pairing of D with F. There were also some instances where no totals were given which lost students a significant number of marks. Students should be advised to be thorough when checking the shortest route between each odd pairing. Many students did not explicitly state the arcs that should be repeated instead stating that DE and FK should be repeated instead of the correct arcs DA, AE, FJ, JK. Furthermore, a number of students did not state the length of a shortest inspection route.

Only a minority of students were able to answer question (b) correctly with the majority stating that J would appear 6 times (6 was the order of vertex J once the two additional arcs were added) rather than the correct answer of 3 times.

In question (c) many students identified DE, DF and EF as the paths that needed to be considered, although they often missed stating the fact that DF was the shortest path that did not including vertex K. Many students, even with the correct selection of arcs, either did not state a route or gave an incorrect route. Some students misunderstood the reasoning altogether and focused on the fact that DK had the greatest weight of the previous pairings and therefore should be avoided and so EF should be repeated. Others still said that 'FK is the least therefore start and end at F and K'. Those that did state DF usually went on to score the mark for stating the length of a shortest inspection route.

Question Number	Scheme	Marks
(a)	D(A)E + F(J)K = 35 + 15 = 50* D(HJ)F + E(FJ)K = 24 + 40 = 64 D(HJ)K + EF = 33 + 25 = 58 Arcs DA, AE, FJ, JK will be traversed twice Route length = 451 + 50 = 501 (km)	M1 A1 (2 correct) A1 (3 correct) A1 A1ft **(5)**
(b)	Vertex J would appear 3 times in the shortest inspection route	B1 **(1)**
(c)	We only have to repeat one pair of odd vertices which does not include vertex K (DE = 35, DF = 24, EF = 25) DF is the smallest of the three so repeat DF (DH, HJ, JF) and therefore the other hut should be built at E Route e.g. EADEHDHJFBEFCGFJHLGKJLMK The length of the route is 475 (km)	DM1 A1 A1 A1ft **(4)** **10 marks**

Notes for Question

a1M1: Three distinct pairings of the **correct** four odd nodes.
a1A1: **Any** two rows correct including pairings **and** totals.
a2A1: **All** three rows correct including pairings **and** totals.
a3A1: CAO correct **arcs** clearly (**not** just in their working) stated: DA, AE, FJ, JK. Accept DAE, FJK or DE via A, FK via J. Do not accept DE, FK.
a4A1ft: The correct answer of 501 **or** 451 + their smallest repeat out of a choice of at least **two** totals seen.

b1B1: CAO (3)

c1DM1: Identifies the need to repeat one path of the three (DE, DF, EF) which does not include K (maybe implicit) or listing of possible repeats – this mark is dependent on scoring the M mark in (a). **Stating any path (DE, DF, EF) that does not include K is sufficient for this mark.**
c1A1: Identifies DF **as the least of those paths not including K and E** as the position of the other hut. They have to explicitly state that DF is the **least** path that does **not include K or** they can list all three paths (DE, DF, EF) and then say DF is the smallest as this implicitly implies that they are considering only paths that do not include K.
c2A1: Any correct route - checks: starts at E and finishes at K (or vice-versa), 24 vertices (D, G, K, L appear twice and E, F, H, J appear three times and every other letter appears once).
c3A1ft: Correct answer of 475 **or** 451+ their DF (i.e. the least path that does not include K – so their smallest of DE, DF or EF – must be **their** smallest value (usually from (a)) not what they state/think is their smallest value).

Q4.

[*The total weight of the network is 384*]

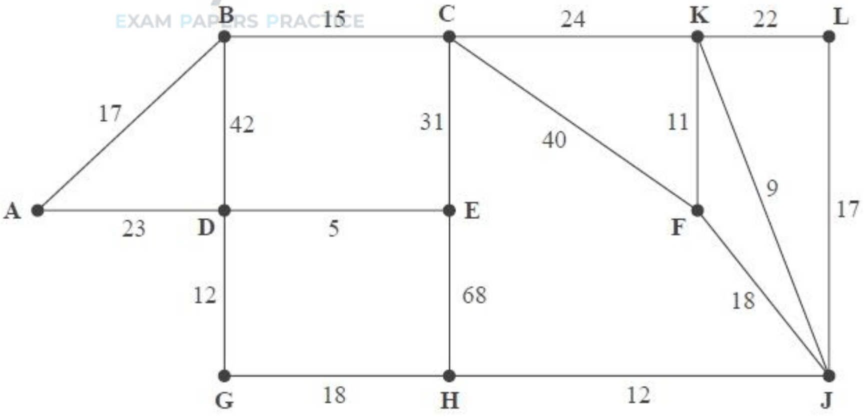


Figure 5

Figure 5 models a network of corridors in an office complex that need to be inspected by a security guard. The number on each arc is the length, in metres, of the corresponding section of corridor.

Each corridor must be traversed at least once and the length of the inspection route must be minimised. The guard must start and finish at vertex A.

(a) Use the route inspection algorithm to find the length of the shortest inspection route. State the arcs that should be repeated. You should make your method and working clear.

(5)

It is now possible for the guard to start at one vertex and finish at a different vertex. An inspection route that traverses each corridor at least once is still required.

(b) Explain why the inspection route should start at a vertex with odd degree.

(2)

The guard decides to start the inspection route at F and the length of the inspection route must still be minimised.

(c) Determine where the guard should finish. You must give reasons for your answer.

(2)

(d) State a possible route and its length.

(2)

Part (a) was generally answered well by most students with the vast majority stating the correct three distinct pairings of the correct four odd nodes. There were a few students who only gave two pairings of the four odd nodes or who gave several pairings but not three distinct pairings. There were however many instances where the totals were incorrect. The majority of such mistakes occurred for the two pairings of BF with EH and BH with EF. There were also some instances where no totals were given which lost students a significant number of marks. Students should be advised to be thorough when checking the shortest route between each odd pairing. Many students did not explicitly state the arcs that should be repeated instead stating that BE and FH should be repeated instead of the correct arcs BA, AD, DE, FJ and JH.

Part (b) was challenging for most students and it was rare to see both marks awarded. The majority of students obtained no marks here due to arguments along the lines of "because that is the method that gives the shortest route" or fairly long-winded arguments to do with the order of nodes and the number of times a node can be entered/left without ever reaching the crux of the correct explanation. Of the two marks available a small minority scored a mark for conveying at least the idea of finishing at an odd vertex.

In part (c), many students identified BE, EH and BH as the paths that needed to be considered, although they often missed stating the fact that EH was the shortest path that did not including vertex F. Many students, even with the correct selection of arcs, either did not state a route or gave an incorrect route in part (d). Some students misunderstood the reasoning altogether and focused on the fact that EF had the greatest weight of the previous pairings and therefore should be avoided and so BH should be repeated. Those that did state EH usually went on to score the mark in part (d) for stating the length of a shortest inspection route.

Question Number	Scheme	Marks
(a)	B(AD)E + F(J)H = 45 + 30 = 75* B(CK)F + E(DG)H = 50 + 35 = 85 B(CKJ)H + E(DGHJ)F = 60 + 65 =125 Arcs BA, AD, DE, FJ and JH will be traversed twice Route length = 384 + 75 = 459 (metres)	M1 A1 (2 correct) A1 (3 correct) A1 A1ft (5)
(b)	e.g. if we start at an odd vertex we will finish at another odd vertex. This removes the need to repeat the route between them. So we just have to consider one repeated route rather than two	B2, 1, 0 (2)
(c)	We only have to repeat one pair of odd vertices which does not include F (BE = 45, EH = 35, BH = 60) EH is the smallest of the repeat so repeat EH (ED, DG, GH) and therefore the guard should finish at B	M1 A1 (2)
(d)	Route e.g. FJKFCKLJHGHEDGDECBDAB The length of the route is 419 (metres)	B1 B1ft (2) **11 marks**

Notes for Question

a1M1: Three distinct pairings of the **correct** four odd nodes
a1A1: **Any** two rows correct including pairings **and** totals
a2A1: **All** three rows correct including pairings **and** totals
a3A1: CAO correct **arcs** clearly (not just in their working) stated: BA, AD, DE, FJ, JH. Accept BADE, FJH or BE via A and D, FH via J. Do not accept BE, FH
a4A1ft: Correct answer of 459, or 384 + their smallest repeat out of a choice of at least **two** totals seen

b1B1: One of (i) **finishing** at an odd vertex (ii) only having to **repeat one route/pairing/pair/path** (but not 'repeat only one arc') rather than two **or** having **one less route/pairing/pair/path to repeat** (but not an argument based only on arcs e.g. 'one less arc to repeat' or 'it reduces the number of arcs')
b2B1: Correct complete argument – including both (i) and (ii) from b1B1 (so B0B1 is not possible in (b))

c1M1: Identifies the need to repeat one route of BE(45), EH(35), BH(60), which does not include F (maybe implicit) or a general comment to repeat one route that does not include F
c1A1: Identifies EH (but not just 35) as the least of those paths not including F, **and** B as the position of the finishing vertex. Note that they must either explicitly state that EH **is the least not including F** (just stating that EH is the least is A0) or they list the three pairings (BE, EH, BH) **and** only these three pairings in this part and state that EH is the least

d1B1: Any correct route – checks: start at F and finishes at B, 21 vertices (repeats ED, DG, GH, and node A appears 1, B(2), C(2), D(3), E(2), F(2), G(2), H(2), J(2), K(2), L(1))

d2B1ft: Correct answer of 419 or 384 + their EH (i.e. the least route that does not include F – so their smallest of BE, EH, BH – must be their smallest value (usually from (a)) not what they state/think is their smallest value). **This mark is dependent on the M mark in (a)**